

Yanshu Shingala

AI / ML & FULL-STACK DEVELOPER

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EDUCATION

JG University B.Tech in Data Science & Artificial Intelligence	Sep 2025 – Present
Marwadi University Diploma in Computer Engineering — CGPA: 6.97	Sep 2022 – May 2025
• Runners-Up, Code Storm inter-college coding competition	

PROJECTS

LAIKA — Local AI Knowledge Assistant	2025
• Built a fully offline RAG (Retrieval-Augmented Generation) system that lets users query their own documents (PDF, DOCX, TXT, MD) with zero data leaving the machine.	
• Implemented hybrid search combining ChromaDB semantic embeddings and SQLite FTS5 full-text search via Reciprocal Rank Fusion (RRF), improving retrieval relevance over single-method approaches.	
• Architected a FastAPI backend + Streamlit frontend with multi-turn conversation memory, citation display, and full Docker Compose deployment support.	
• Integrated local Ollama LLMs (qwen2.5:3b) for inference — no API keys or internet required.	
PythonFastAPIChromaDBSQLite FTS5StreamlitOllamaDockerRAG	

Vaani — Offline Multilingual Voice Assistant	2025
• Engineered a 100% offline voice assistant supporting Hindi, Gujarati, and English — no API keys, no cloud, all audio stays on-device.	
• Built a 5-stage AI pipeline : WebRTC noise cancellation → Faster-Whisper STT → Meta NLLB-200 translation → Ollama LLM → offline SAPI5 TTS.	
• Implemented real-time silence detection and background-threaded TTS so the assistant feels responsive with no blocking waits.	
PythonFaster-WhisperMeta NLLB-200OllamaWebRTC VADStreamlitNLP	

House Price Predictor	2025 · Deployed on Render
• Trained a scikit-learn regression model on the Ames Housing dataset achieving R² = 0.91, MAE = \$15,035 on held-out test data.	
• Built and deployed a Flask REST API with a /predict endpoint, full input validation, and Gunicorn production server via Procfile — live on Render.	
Pythonscikit-learnFlaskPandasNumPyGunicornRender	

Email Spam Detector	2024
• Built a full-stack ML web app using TF-IDF vectorization + logistic regression to classify emails as spam or legitimate with a live web interface.	
Pythonscikit-learnTF-IDFVercel	

SKILLS

Languages	Python, SQL, JavaScript, C++, HTML/CSS
AI / ML	scikit-learn, Faster-Whisper, Meta NLLB-200, Ollama, RAG, TF-IDF, Prompt Engineering
Backend	Flask, FastAPI, REST APIs, Gunicorn
Frontend	React, Streamlit, Tailwind CSS
Databases	ChromaDB, SQLite, PostgreSQL
Tools	Git, Docker, Vercel, Render, Google Cloud

CERTIFICATIONS — GOOGLE CLOUD (ARCADE PROGRAM)

Create ML Models with BigQuery ML	Credly ↗
Prepare Data for ML APIs on Google Cloud	Credly ↗
Store, Process, and Manage Data on Google Cloud — Console	Credly ↗
Use Machine Learning APIs on Google Cloud	Credly ↗

ACHIEVEMENTS

Runners-Up — Code Storm, Marwadi University Inter-college competitive programming competition	2024
Microsoft Copilot for Productivity — Microsoft & LinkedIn Learning Applied AI productivity tools across M365 (Word, Excel, PowerPoint, Outlook, Teams)	